

ECP Calculation Report (Packed Column — Stage C4)

Test

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This document was generated from a frozen report payload linked to Design Revision Rev 0 (internal id 6) of design LLX-RND-2026-0001. It does not render live database values; regeneration from the same payload reproduces this document exactly.

Calculation runs consumed:

- llx-ecp 1.0.0 — run #63 (2026-08-06 11:25 UTC)

Renders the persisted 'ecp' result snapshot (computed 2026-08-06 11:25 UTC). Applicability: PRELIMINARY ECP-TYPE PACKED COLUMN SCREENING — NOT VENDOR RATING AND NOT FOR FABRICATION

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1. Introduction, Design Basis & Run Identification

This ECP Calculation Report presents the Stage C4 packed-column screening for design LLX-RND-2026-0001, Test, rendered from the frozen 'ecp' result snapshot of Design Revision Rev 0. Calculation run: #63 (engine llx-ecp v1.0.0, status 'warning', calculated 2026-08-06 11:25 UTC). The engine was NOT re-run for this report.

Engine chain versions (verbatim from snapshot): cel v1.0.0, epd v1.0.0, processDesign v1.0.0, ecpPackedColumn v1.0.0, hydraulicsCommon v1.0.0. The governing design basis is stated in LLX-RND-2026-0001-DBR; process flows derive from LLX-RND-2026-0001-PDR; generic hydraulics from LLX-RND-2026-0001-HDR (same revision).

Engine applicability statement (verbatim): "PRELIMINARY ECP-TYPE PACKED COLUMN SCREENING — NOT VENDOR RATING AND NOT FOR FABRICATION".

2. Design Basis Inputs

Item	Value	Unit	Source Type	Source Reference
Operating temperature	70	°C	Design basis	—
Phase configuration	nmp_continuous_rr bo_dispersed	—	Engineer Entry	Phase continuity is taken from the engineer input only.
Dispersed / continuous phase	RRBO / NMP	—	Engineer Entry	—
Theoretical stages (from C2)	6	—	C2 frozen result	—
Height basis	HETS	—	Engine configuration	—
HETS	0.75	m	Assumed	Thermopac Preliminary ECP Screening Default v1.0 — HETS is SYSTEM data (solvent + feed + packing + temperature), never packing data alone
System derating factor	1	—	Not supplied — default 1.0	Not supplied — 1.0 applied with NO_SYSTEM_DERATING_DATA warning; never invented
RRBO density used	870	kg/m3	Entered/EPD	Assumed: Thermopac Feed Master (Default) — Fluid Properties auto-populated value, no engineer source type selected
NMP density used	986.272727272727 3	kg/m3	Entered/EPD	Source-tagged tabular data (no fitted correlation approved yet) — CRC Press / provisional (2016)
Utilization screening band	40 – 80	%	Configurable criterion	Configurable screening criterion — not a universal engineering rule

3. Packing Data (Consumed Record — Verbatim)

Packing Database record 'sulzer-smvp-preliminary-screening'. The engine consumes packing data — it never owns packing data. Vendor identity is data, not code.

Item	Value	Unit	Source Type	Source Reference
Record ID / revision	sulzer-smvp-preliminary-screening / Rev 0 (preliminary screening)	—	Packing Database	—
Product	SMVP — Preliminary Screening Record (Structured LLX packing)	—	Packing Database	Sulzer (preliminary screening record — not vendor-certified)
Type / geometry class / material	structured sheet-metal / structured / Metal (stainless steel assumed for screening)	—	Packing Database	—
Nominal size	NOT ENTERED	—	—	—

Item	Value	Unit	Source Type	Source Reference
Void fraction	0.95	-	Assumed	Generic structured sheet-metal packing literature-typical value, Assumed — NOT Sulzer-certified vendor data
Specific surface area	250	m ² /m ³	Assumed	Packing designation: "250" is the nominal specific-surface-area grade (250 m ² /m ³) of the SMV/SMVP families, not a physical element dimension; within the published 200–500 m ² /m ³ SSA screening range (Johannes Rauber, Sulzer Chemtech Ltd., Design Practice for Packed Liquid-Liquid Extraction Columns, AIChE 2006 — preliminary screening characteristics only)
Record source	—	—	Controlled literature	Johannes Rauber, Sulzer Chemtech Ltd., Design Practice for Packed Liquid-Liquid Extraction Columns, AIChE 2006 — preliminary screening characteristics only
Vendor notes	—	—	Packing Database	Efficiency / Back-Mixing Preferred family (published screening throughput 35–60 m ³ / (m ² ·h), NTS ≤ 10, up to 3 beds). Preliminary screening record — controlled literature characteristics (Johannes Rauber, Sulzer Chemtech Ltd., Design Practice for Packed Liquid-Liquid Extraction Columns, AIChE 2006 — preliminary screening characteristics only). Replace with a vendor-certified record before any rating use.

4. Equation Reference Index (From Frozen Snapshot)

Every formula/basis string stored in the frozen snapshot, grouped by equation reference and reproduced verbatim. The controlled statements live in the C4 section of the Correlation & Equation Register; software-verification status is tracked in the V&V equation register (all ECP entries currently Partially Verified).

Equation Ref	Formula / basis strings as stored in the frozen snapshot (verbatim; one line per distinct usage)
ECP-001	A = $\pi \cdot D^2/4$ Total both-phase liquid load Q_RRBO/A Q_NMP/A
ECP-002	Vendor Packing Capacity
ECP-003	Vendor minimum wetting rate Vendor recommended loading range
ECP-005	theoreticalStages (engineer) × HETS (Assumed: Thermopac Preliminary ECP Screening Default v1.0; system: RRBO (Re-Refined Base Oil)/NMP on SMVP — Preliminary Screening Record at 70 °C)
ECP-007	Vendor Pressure Drop (wet)
ECP-008	Assumed: Thermopac Preliminary Equipment Screening Default v1.0 ECP-005/ECP-006 (1 bed(s)) Total T/T + top head + bottom head Sum of tangent-to-tangent items (heads excluded)

5. HETS & Packing Height

HETS provenance (verbatim): value 0.75 m, source type Assumed, reference "Thermopac Preliminary ECP Screening Default v1.0"; system: RRBO (Re-Refined Base Oil) / NMP on SMVP — Preliminary Screening Record at 70 °C. Note (verbatim): HETS is SYSTEM data (solvent + feed + packing + temperature), never packing data alone.

Item	Value	Unit	Source Type	Source Reference
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Item	Value	Unit	Source Type	Source Reference
Packed height (N × HETS) [ECP-005]	4.50	m	Pending Validation	theoreticalStages (engineer) × HETS (Assumed: Thermopac Preliminary ECP Screening Default v1.0; system: RRBO (Re- Refined Base Oil)/NMP on SMVP — Preliminary Screening Record at 70 °C)

6. Bed Arrangement

Basis (verbatim): No vendor bed-height limit supplied — single bed for screening (NO_BED_HEIGHT_LIMIT_DATA)
[ECP-006]

Bed	Height (m)
1	4.50
Redistributors	0

7. Case Flows — Normal

Item	Value	Unit	Source Type	Source Reference
NMP (continuous) mass flow	5917.64	kg/h	From C2 frozen flows	—
RRBO (dispersed) mass flow	3480.00	kg/h	From C2 frozen flows	—
NMP volumetric flow	6.00	m³/h	From C2 frozen flows	—
RRBO volumetric flow	4.00	m³/h	From C2 frozen flows	—
Utilization screening band	40 – 80	%	Configurable criterion	Configurable screening criterion — not a universal engineering rule

8. Hydraulic Loading & Capacity Screening — Normal Case

Per-diameter hydraulic loading (ECP-001), utilization vs Vendor Packing Capacity (ECP-002 — not screenable for this record), vendor checks (ECP-003) and wet pressure drop (ECP-007). "Not Calculable" cells reproduce the engine finding — no value was invented.

D (m)	Area (m²)	u_c NMP (m³/m²·h)	u_d RRBO (m³/ m²·h)	Total load (m³/m²·h)	Utilization (%)	DeltaP (Pa)	Min wetting	Rec. loading	Feasibility
0.30	0.071	84.88	56.59	141.47	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.35	0.096	62.36	41.58	103.94	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.40	0.126	47.75	31.83	79.58	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.45	0.159	37.73	25.15	62.88	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.50	0.196	30.56	20.37	50.93	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.55	0.238	25.25	16.84	42.09	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.60	0.283	21.22	14.15	35.37	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.65	0.332	18.08	12.05	30.14	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.70	0.385	15.59	10.39	25.98	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.75	0.442	13.58	9.05	22.64	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation
0.80	0.503	11.94	7.96	19.89	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_va lilation

D (m)	Area (m²)	u_c NMP (m³/m²·h)	u_d RRBO (m³/m²·h)	Total load (m³/m²·h)	Utilization (%)	DeltaP (Pa)	Min wetting	Rec. loading	Feasibility
0.85	0.567	10.57	7.05	17.62	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.90	0.636	9.43	6.29	15.72	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.95	0.709	8.46	5.64	14.11	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.00	0.785	7.64	5.09	12.73	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.05	0.866	6.93	4.62	11.55	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.10	0.950	6.31	4.21	10.52	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.15	1.039	5.78	3.85	9.63	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.20	1.131	5.31	3.54	8.84	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.25	1.227	4.89	3.26	8.15	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.30	1.327	4.52	3.01	7.53	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.35	1.431	4.19	2.79	6.99	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.40	1.539	3.90	2.60	6.50	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.45	1.651	3.63	2.42	6.06	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.50	1.767	3.40	2.26	5.66	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.55	1.887	3.18	2.12	5.30	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.60	2.011	2.98	1.99	4.97	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.65	2.138	2.81	1.87	4.68	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.70	2.270	2.64	1.76	4.41	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.75	2.405	2.49	1.66	4.16	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.80	2.545	2.36	1.57	3.93	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.85	2.688	2.23	1.49	3.72	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.90	2.835	2.12	1.41	3.53	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.95	2.986	2.01	1.34	3.35	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
2.00	3.142	1.91	1.27	3.18	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation

9. Case Flows — Maximum Continuous

Item	Value	Unit	Source Type	Source Reference
NMP (continuous) mass flow	7101.16	kg/h	From C2 frozen flows	—
RRBO (dispersed) mass flow	3480.00	kg/h	From C2 frozen flows	—

Item	Value	Unit	Source Type	Source Reference
NMP volumetric flow	7.20	m³/h	From C2 frozen flows	—
RRBO volumetric flow	4.00	m³/h	From C2 frozen flows	—
Utilization screening band	40 – 80	%	Configurable criterion	Configurable screening criterion — not a universal engineering rule

10. Hydraulic Loading & Capacity Screening — Maximum Continuous Case

D (m)	Area (m²)	u_c NMP (m³/m²·h)	u_d RRBO (m³/m²·h)	Total load (m³/m²·h)	Utilization (%)	DeltaP (Pa)	Min wetting	Rec. loading	Feasibility
0.30	0.071	101.86	56.59	158.45	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.35	0.096	74.84	41.58	116.41	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.40	0.126	57.30	31.83	89.13	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.45	0.159	45.27	25.15	70.42	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.50	0.196	36.67	20.37	57.04	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.55	0.238	30.31	16.84	47.14	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.60	0.283	25.46	14.15	39.61	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.65	0.332	21.70	12.05	33.75	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.70	0.385	18.71	10.39	29.10	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.75	0.442	16.30	9.05	25.35	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.80	0.503	14.32	7.96	22.28	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.85	0.567	12.69	7.05	19.74	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.90	0.636	11.32	6.29	17.61	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
0.95	0.709	10.16	5.64	15.80	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.00	0.785	9.17	5.09	14.26	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.05	0.866	8.32	4.62	12.93	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.10	0.950	7.58	4.21	11.79	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.15	1.039	6.93	3.85	10.78	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.20	1.131	6.37	3.54	9.90	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.25	1.227	5.87	3.26	9.13	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.30	1.327	5.42	3.01	8.44	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.35	1.431	5.03	2.79	7.82	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.40	1.539	4.68	2.60	7.28	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation

D (m)	Area (m ²)	u_c NMP (m ³ /m ² ·h)	u_d RRBO (m ³ /m ² ·h)	Total load (m ³ /m ² ·h)	Utilization (%)	DeltaP (Pa)	Min wetting	Rec. loading	Feasibility
1.45	1.651	4.36	2.42	6.78	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.50	1.767	4.07	2.26	6.34	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.55	1.887	3.82	2.12	5.94	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.60	2.011	3.58	1.99	5.57	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.65	2.138	3.37	1.87	5.24	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.70	2.270	3.17	1.76	4.93	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.75	2.405	2.99	1.66	4.66	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.80	2.545	2.83	1.57	4.40	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.85	2.688	2.68	1.49	4.17	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.90	2.835	2.54	1.41	3.95	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
1.95	2.986	2.41	1.34	3.75	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation
2.00	3.142	2.29	1.27	3.57	Pending Validation	Not Calculable	Not Calculable	Not Calculable	pending_validation

11. Capacity / Flooding & Pressure-Drop Architecture

Capacity/flooding screening: hydraulic utilization is defined against Vendor Packing Capacity (ECP-002). The consumed packing record carries no Vendor Packing Capacity data, therefore utilization is Pending Validation at every diameter and the 40–80 % screening band could not be applied. The C3 generic throughput percentage is not a substitute (engine statement, verbatim).

Pressure-drop architecture (verbatim): Dry and wet pressure drop are separated; only WET applies to the operating extraction column. Dry basis is reserved architecture. (wetApplied=true, dryReserved=true).

Rate-based placeholders (verbatim): Reserved architecture for a future rate-based path (heightBasis HTU_NTU). NOT calculated in Stage C4. (HTU/NTU/ka/interfacial area all null — reserved, not calculated).

12. Height Breakdown — Disengagement, T/T & Overall

Internals stack from the frozen snapshot (ECP-008). Disengagement allowances are the Top/Bottom Disengagement lines.

Item	Value	Unit	Source Type	Source Reference
Top Head [ECP-008]	0.50	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Top Disengagement [ECP-008]	1.00	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Top Distributor [ECP-008]	0.50	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Packing Bed 1 [ECP-008]	4.50	m	Pending Validation	ECP-005/ECP-006 (1 bed(s))
Hold-Down [ECP-008]	0.20	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Packing Support [ECP-008]	0.20	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Bottom Distributor [ECP-008]	0.50	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0

Item	Value	Unit	Source Type	Source Reference
Bottom Disengagement [ECP-008]	1.00	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Bottom Head [ECP-008]	0.50	m	Pending Validation	Assumed: Thermopac Preliminary Equipment Screening Default v1.0
Total tangent-to-tangent (T/T) [ECP-008]	7.90	m	Pending Validation	Sum of tangent-to-tangent items (heads excluded)
Overall vessel height [ECP-008]	8.90	m	Pending Validation	Total T/T + top head + bottom head

13. Technology Comparison — ECP vs ECR (Report-Layer Derivation)

This side-by-side table is derived MECHANICALLY in the report layer from the two frozen C4/C5 snapshots of the same design revision. It is not an engine result and it expresses no technology preference — both technologies remain Pending Validation and neither utilization band can be screened without vendor capacity data.

Item	ECP (Packed Column, C4)	ECR (Agitated Column, C5)
Theoretical stages (from C2)	6	6
Stage device	HETS 0.75 m (Assumed)	15 compartments × 0.25 m (efficiency 0.4, Assumed)
Active height	4.50 m packed (Pending Validation)	3.75 m agitated (Pending Validation)
Total tangent-to-tangent	7.90 m	6.75 m
Overall vessel height	8.90 m	8.75 m
Internals basis	Packing record: SMVP — Preliminary Screening Record (Rev 0 (preliminary screening)) — Assumed screening data, not vendor-certified	Rotor: Kühni turbine (default label); D_R/D ratio 0.5 (Assumed)
Mechanical drive	None (static internals)	Agitated — motor design power is diameter- and speed-dependent (see ECR power tables); screening only, not a motor selection
Hydraulic utilization screening	Not screenable — no Vendor Packing Capacity data in the packing record	Not screenable — no ECR-specific Vendor Hydraulic Capacity data
Pressure drop	Not Calculable — no wet pressure-drop basis in the packing record	Not emitted by the C5 screening scope
Run status	pending_validation	pending_validation

14. Engine Warnings (Verbatim)

No warning strings stored on individual items. Run status: pending_validation — see Validation & Missing-Data Summary.

15. Limitations (Verbatim)

1. No proprietary packing model
2. No vendor hydraulic guarantee
3. HETS requires vendor/test confirmation
4. No droplet breakup/coalescence model
5. No axial-dispersion model
6. No rate-based mass-transfer model
7. No mechanical code design

16. Independent-Verifiability Statement

Items in the table below are NOT independently verifiable from this report: the engine declared them Not Calculable because the governing vendor/laboratory datum is absent, and no value was invented. A checker cannot reproduce a number that was never computed — closure of these items requires the named data, not re-calculation.

In addition, every ECP result carrying status 'Pending Validation' is ARITHMETICALLY verifiable from the inputs and

formula strings in this report, but NOT engineering-verifiable: its basis contains Assumed data (see the Assumptions Register). Hand-checking confirms the arithmetic only; it does not validate the assumed values.

Equation Ref	Item (stored basis string)	Why it cannot be independently verified from this report (stored explanation, verbatim)
ECP-004	—	No distributor specification supplied — checks run only when vendor distributor data exist
ECP-007	Vendor Pressure Drop (wet)	No wet pressure-drop basis in the packing record — no universal Pa/m value is invented; all other outputs are unaffected
ECP-003	Vendor minimum wetting rate	No vendor minimum wetting rate in the packing record — never assumed
ECP-003	Vendor recommended loading range	No vendor recommended loading range in the packing record

17. Assumptions Register

Every value below carries source type Assumed or a Thermopac screening default and is Pending Validation. This register is extracted mechanically from the tagged source data — it cannot be edited in the report layer.

Item	Value	Source Reference	Validation Status
Packing record 'sulzer-smvp-preliminary-screening' contains Assumed data	—	All packing-derived items are Pending Validation until vendor/measured data replace the Assumed entries	Assumed — Pending Validation
Fluid property data contain Assumed entries (RRBO entered tags and/or EPD data)	—	All load-derived items (ECP-001 loads, ECP-002 utilization) are Pending Validation	Assumed — Pending Validation
HETS = 0.75 m is ASSUMED (Thermopac Preliminary ECP Screening Default v1.0)	—	Packing height and total heights are Pending Validation	Assumed — Pending Validation
Top Head allowance 0.5 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Top Disengagement allowance 1 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Top Distributor allowance 0.5 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Packing Support allowance 0.2 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Hold-Down allowance 0.2 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Bottom Distributor allowance 0.5 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Bottom Disengagement allowance 1 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
Bottom Head allowance 0.5 m is ASSUMED	—	Height breakdown is Pending Validation	Assumed — Pending Validation
HETS	0.75 m	Thermopac Preliminary ECP Screening Default v1.0	Assumed — Pending Validation
Packing void fraction	0.95 -	Generic structured sheet-metal packing literature-typical value, Assumed — NOT Sulzer-certified vendor data	Assumed — Pending Validation
Packing specific surface area	250 m ² /m ³	Packing designation: "250" is the nominal specific-surface-area grade (250 m ² /m ³) of the SMV/SMVP families, not a physical element dimension; within the published 200–500 m ² /m ³ SSA screening range (Johannes Rauber, Sulzer Chemtech Ltd., Design Practice for Packed Liquid-Liquid Extraction Columns, AIChE 2006 — preliminary screening characteristics only)	Assumed — Pending Validation

18. Validation & Missing-Data Summary

Severity	Item	Reason
WARNING	Vendor Packing Capacity	No Vendor Packing Capacity data in the packing record — hydraulic utilization cannot be screened; the C3 generic percentage is not a substitute (ECP-002).
WARNING	Vendor wet pressure-drop basis	No wet pressure-drop basis in the packing record — pressure drop Not Calculable at every diameter; no universal Pa/m value is invented (ECP-007).
WARNING	Vendor minimum wetting rate	Not in the packing record — minimum-wetting check Not Calculable; never assumed (ECP-003).
WARNING	Vendor recommended loading range	Not in the packing record — recommended-loading check Not Calculable (ECP-003).
WARNING	Distributor specification	No distributor specification supplied — distributor checks run only when vendor distributor data exist (ECP-004).
WARNING	Vendor-certified packing record	Packing record 'sulzer-smvp-preliminary-screening' is a preliminary screening record with Assumed characteristics — replace with a vendor-certified record before any rating use.
WARNING	HETS confirmation	HETS is Assumed (system data: solvent + feed + packing + temperature) — vendor/test confirmation required before the packed height is engineering-verified (ECP-005).
WARNING	System derating factor	Not supplied — 1.0 applied with NO_SYSTEM_DERATING_DATA warning; never invented.

19. References

- [1] Johannes Rauber, Sulzer Chemtech Ltd., Design Practice for Packed Liquid-Liquid Extraction Columns, AIChE 2006 — preliminary screening characteristics only
[2] Packing Database record 'sulzer-smvp-preliminary-screening'

20. Revision History

Report Rev	Date	Generated By	Description
Rev 0	2026-08-06	Manager	Superseded generation
Rev 0	2026-08-06	Manager	Regenerated from current design revision data

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